

Total No. of Questions : 9]

SEAT No. :

PE-896

[Total No. of Pages : 4

[6581]-1902
F.E. (Common)
ENGINEERING PHYSICS
(2019 Pattern) (Semester - I/II) (107002)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Q. 1 is compulsory, Answer Q2 or Q3, Q4 or Q5, Q6 or Q7, Q8 or Q9.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicates full marks.
- 4) Use of Electronic pocket calculator is allowed.
- 5) Assume suitable data, if necessary.

Physical Constants: -

- Plank's constant $h = 6.63 \times 10^{-34} \text{ J.S}$
- Mass of electron $m_e = 9.1 \times 10^{-31} \text{ Kg}$
- Charge on electron $e = 1.6 \times 10^{-19} \text{ C}$

Q1) Write the correct option with answer for the following (1 mark each) : **[10]**

- i) Which of the following is the basic unit of information in quantum computing?
 - a) Bit
 - b) Qubit
 - c) Byte
 - d) Terabyte
- ii) Which pair of variables is applicable to Heisenberg's uncertainty principle?
 - a) wavelength and energy
 - b) position and momentum
 - c) charge and displacement
 - d) atomic radius and time
- iii) The concept of matter wave was suggested by
 - a) Heisenberg
 - b) de Broglie
 - c) Schrodinger
 - d) Laplace

P.T.O.

- iv) Which is the correct ordering of the band gap energy?
- a) Diamond > silicon > copper b) Diamond < silicon < copper
c) Diamond < silicon > copper d) Diamond = silicon < copper
- v) Hall effect is true for
- a) P-type semiconductors only
b) N-type Semiconductors only
c) metals only
d) both for Metals and semiconductors
- vi) In the ferromagnetic materials, relative permeability is ____.
- a) Less than 1 b) Just greater than 1
c) Significantly greater than 1 d) None of these
- vii) In a Josephson junction, two superconductors are joined by a _____ of a few angstroms thickness.
- a) Superconductor material b) Insulating material
c) Conducting material d) Magnetic material
- viii) Below _____ temperature, superconducting material shows zero resistivity.
- a) Inversion b) Conversion
c) Critical d) Superconducting
- ix) A quantum wire has _____ dimensions less than 100 nm size.
- a) 0 b) 1
c) 2 d) 3
- x) X - ray or gamma rays are passed through a material and their absorption and scattering are analysed in
- a) Acoustic emission technique
b) Ultrasonic testing
c) Radiographic techniques
d) All of the above

- Q2)** a) Starting from de Broglie wave equation, derive Schrodinger's Time Independent wave equation. [6]
- b) State de Broglie hypothesis. Derive the equation of de Broglie wavelength in terms of Kinetic Energy of a particle. [5]
- c) A neutron is trapped in infinite potential well of width 10^{-14} m. Calculate its first energy eigen value in eV. (Mass of neutron = 1.675×10^{-27} kg) [4]

OR

- Q3)** a) Derive the equation for the energy of particle is enclosed in a one-dimensional rigid box. [6]
- b) What is Heisenberg's uncertainty Principle? Obtain Heisenberg's uncertainty relation in terms of energy and time. [5]
- c) Calculate the De-Broglie wavelength of an electron accelerated through potential difference of 10,000 V. [4]

- Q4)** a) Derive an expression for the conductivity in an intrinsic and extrinsic Semiconductor. [6]
- b) Define a Fermi level in semiconductor. Using Fermi-Dirac probability distribution function, show that the position of Fermi energy level in the intrinsic semiconductor is midway of the forbidden energy gap. [5]
- c) Calculate the mobility of charge carriers in doped silicon whose conductivity is $100 (\Omega\text{-m})^{-1}$ and the Hall coefficient is $3.6 \times 10^{-4} \text{ m}^3/\text{coulomb}$. [4]

OR

- Q5)** a) State and Explain Hall Effect. Derive the equation of Hall voltage and Hall coefficient. [6]
- b) What is photovoltaic effect? Explain the construction and working of solar cell. [5]
- c) A sample of intrinsic germanium at room temperature has a carrier concentration of $4.41 \times 10^{22} \text{ cm}^{-3}$. Donor impurity is added in the ratio 1 donor atom per 10^8 atoms/ cm^3 . Determine the resistivity of the material. [mobility of electrons is $3800 \text{ cm}^2/\text{V. s}$] [4]

- Q6)** a) Explain the process of data is recording, retrieving and erase in magneto-optical recording devices. [6]
- b) State and explain Meissner effect. Show that superconductors are perfect diamagnetic in nature. [5]
- c) Calculate the critical magnetic field of wire of lead having a diameter of 1 mm at 4.2 K. The critical temperature for lead is 7.18 K and critical magnetic field at 0K is 6.5×10^4 A/m. [4]

OR

- Q7)** a) State and explain the term superconductivity. Differentiate between type I and type II superconductors (any 04 points). [6]
- b) Define the terms Magnetic Susceptibility, Relative Permeability and obtain relation between them. [5]
- c) Define the terms and give its SI units : [4]
- i) Magnetic field strength (H)
- ii) Magnetic flux density (B)

- Q8)** a) State various types of non-destructive techniques. Explain Acoustic Emission Technique in brief with applications. [6]
- b) What is nanotechnology? Explain how quantum confinement and change in surface area to volume ratio affects the properties of nanoparticles. [5]
- c) State different application of Nanotechnology and Explain applications of nanoparticles in targeted drug delivery. [4]

OR

- Q9)** a) Explain Optical and electrical properties of nanoparticles in details. [6]
- b) What is NDT? Differentiate between destructive testing and non-destructive testing (any 04 points). [5]
- c) An ultrasonic pulse of frequency 130 kHz is sent through a block of steel. The echo pulse is received after 1.695 microseconds. If velocity of ultrasonic wave in steel is 5900 m/s, calculate the thickness of the steel block and the wavelength of the pulse. [4]

